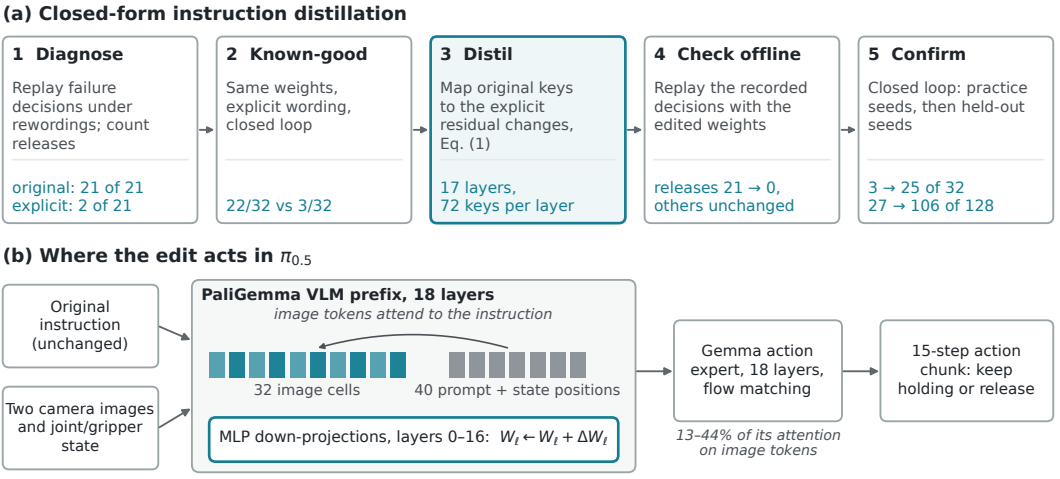
(a) Closed-form instruction distillation
1 Diagnose
Replay failure decisions under rewordings; count releases
original: 21 of 21
explicit: 2 of 21
2 Known-good
Same weights, explicit wording, closed loop
22/32 vs 3/32
3 Distil
Map original keys to the explicit residual changes, Eq. (1)
17 layers, 72 keys per layer
4 Check offline
Replay the recorded decisions with the edited weights
releases 21 → 0, others unchanged
5 Confirm
Closed loop: practice seeds, then held-out seeds
3 → 25 of 32
27 → 106 of 128
(b) Where the edit acts in $\pi_{0.5}$
Original instruction (unchanged)
Two camera images and joint/gripper state
PaliGemma VLM prefix, 18 layers
image tokens attend to the instruction
32 image cells
40 prompt + state positions
MLP down-projections, layers 0–16: $W_\ell \leftarrow W_\ell + \Delta W_\ell$
Gemma action expert, 18 layers, flow matching
13–44% of its attention on image tokens
15-step action chunk: keep holding or release